Institutional Signal Simplification under Stochastic Volatility

With an Application to U.S. Corporate Credit Spreads

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Abstract

Institutions — central banks, rating agencies, regulators, index providers — rarely publish the full state of the systems they observe. Instead they publish a compressed signal obtained by averaging over time (a *horizontal* operator) and over a reference group of peers (a *vertical* operator).

We develop a formal theory of this compression, showing that (i) the two operators generally fail to commute, and the size of the resulting discrepancy is governed by peer turnover and window length; (ii) embedding the signal in a feedback loop with the state it measures yields two structurally distinct instability channels, one from window mismatch and one internal to the volatility process itself; and (iii) the two operators destroy information about the underlying state in qualitatively different ways, with vertical (peer-relative) operators subject to a hard information ceiling that horizontal operators do not share.

We extend the model to continuous-time GARCH-diffusion volatility, derive the joint tail law of the commutator process, and obtain a closed-form expression for time-varying vertical informativeness. We test the framework empirically on Moody's Aaa/Baa corporate bond spreads (FRED, 1986–2026), confirming the theoretical null result under a static peer set and documenting an asymmetry in volatility persistence between systemic and idiosyncratic credit-risk components. We close with a cross-disciplinary discussion spanning economics, statistics, computer science, psychology, and political science.

The paper ends with "The End"

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1 Introduction

Every institution that observes a rich underlying reality and reports a simpler public signal is performing an act of *simplification*: a central bank condenses a multidimensional macroeconomic state into a policy rate and a stance adjective; a rating agency condenses a firm's cash flows, covenants, and industry exposure into a letter grade; an index provider condenses thousands of return series into a single number. This paper asks what structure that simplification has, and in particular what happens when the simplification is built out of two distinct kinds of averaging: averaging along time for a single entity (a *horizontal* rolling window), and averaging across a reference group of peer entities at a fixed time (a *vertical* rolling window), where the peer group itself may change composition over time.

The central technical claims of the paper are:

1. Horizontal and vertical operators generally do not commute, and the size of the non-commutativity (the *commutator*) is a function of peer turnover and horizontal memory length, not of the underlying state itself (Section 3).

2. When the published signal feeds back into the state it measures, the resulting dynamical system has two independent failure modes: a resonance-type instability driven by the relationship between the horizontal and vertical time constants, and an amplitude instability internal to the volatility process (Section 4).

3. The two operators impose different information-theoretic floors: horizontal smoothing can be driven to zero informativeness by lengthening the window, while vertical (peer-demeaning) operators cannot recover systemic information no matter how large the peer group becomes (Section 5).

4. Under continuous-time GARCH-diffusion volatility, these three results interlock: the commutator's tail is a power law whose index is fixed by the same volatility-of-volatility parameter that governs the stability of the volatility process, and vertical informativeness inherits a state-dependent bias whose sign depends on whether volatility is systemic or idiosyncratic (Section 6).

We then test elements of this theory using publicly available data on U.S. corporate bond credit spreads (Section 7), and discuss the theory's connections to adjacent fields (Section 8).

2 Formal Setup

Definition 1 (State and signal)**.** *Let $X_{i,t} \in \mathcal{X}$ denote the true underlying state of entity i at time t. The institution publishes a coarse signal $S_{i,t} \in \mathcal{S}$ with $|\mathcal{S}| \ll |\mathcal{X}|$, obtained as $S_{i,t} = Q(\Omega[X])_{i,t}$, where Ω is a composition of averaging operators and Q is a final quantization map.*

Definition 2 (Horizontal operator)**.** *For weights ϕ and window length w,*

$$H_w[X]_{i,t} = \phi\big(X_{i,t-w+1}, \ldots, X_{i,t}\big).$$

In its linear (equal-weight) form, $H_w[X]_{i,t} = \frac{1}{w}\sum_{s=0}^{w-1} X_{i,t-s}$.

Definition 3 (Vertical operator)**.** *For a peer (reference) set $N_k(i,t)$ of size k, which may change composition over time,*

$$V_k[X]_{i,t} = \psi\big(X_{i,t}, \{X_{j,t} : j \in N_k(i,t)\}\big).$$

In its linear (demeaning) form, $V_k[X]_{i,t} = X_{i,t} - \bar{X}_{N_k(i,t)}(t)$, with $\bar{X}_N(u) = \frac{1}{k}\sum_{j\in N} X_{j,u}$.

Both operators are *garblings* in the sense of [1]: each is a stochastic (or, in the linear case, deterministic linear) reduction of the underlying signal, so that for any composition Ω,

$$I(X;S) \;\le\; I(X;\Omega[X]) \;\le\; \log|\mathcal{S}|,$$

where $I(\cdot;\cdot)$ denotes mutual information. Compression can only destroy information about X; it cannot create it.

3 Non-Commutativity of the Two Operators

Definition 4 (Windowing commutator)**.**

$$C_{i,t} \;=\; \big(H_w V_k - V_k H_w\big)[X]_{i,t}.$$

Proposition 1 (Commutativity under a static peer set)**.** *If $N_k(i,t) = N_k(i)$ is constant in t and both ϕ, ψ are linear, then $C_{i,t} = 0$ for all i, t.*

Proof. With a time-invariant peer set, $\bar{X}_{N_k(i)}(u)$ is a linear functional of $X_{\cdot,u}$ for every u. Then

$$H_w V_k[X]_{i,t} = \frac{1}{w}\sum_{s=0}^{w-1}\Big(X_{i,t-s} - \bar{X}_{N_k(i)}(t-s)\Big) = \frac{1}{w}\sum_{s=0}^{w-1} X_{i,t-s} - \frac{1}{w}\sum_{s=0}^{w-1}\bar{X}_{N_k(i)}(t-s),$$

$$V_k H_w[X]_{i,t} = \frac{1}{w}\sum_{s=0}^{w-1} X_{i,t-s} - \overline{\left(\frac{1}{w}\sum_{s=0}^{w-1} X_{\cdot,t-s}\right)}_{N_k(i)} = \frac{1}{w}\sum_{s=0}^{w-1} X_{i,t-s} - \frac{1}{w}\sum_{s=0}^{w-1}\bar{X}_{N_k(i)}(t-s),$$

where the second equality uses linearity of $\bar{X}_{N_k(i)}(\cdot)$ to exchange the order of summation over s and over $N_k(i)$. The two expressions are identical, hence $C_{i,t} = 0$. □

Proposition 1 is a genuine boundary case, not a triviality: it identifies peer turnover as the *sole* source of non-commutativity when both operators are linear. We verify this numerically in Section 7.

Proposition 2 (Commutator magnitude under peer turnover)**.** *Let peer set membership turn over at rate γ per period (i.e. the expected fraction of a size-k peer set replaced after s periods is $\rho_s \approx \min(1, \gamma s)$), and let X have cross-sectional variance σ_X^2. Then*

$$\mathbb{E}\big[C_{i,t}^2\big] \;=\; \Theta\!\left(\frac{\gamma w \sigma_X^2}{k}\right).$$

Proof sketch. Expanding the two compositions and cancelling the terms that do not involve the peer set leaves

$$C_{i,t} = \frac{1}{w}\sum_{s=0}^{w-1}\Big[\bar{X}_{N_k(i,t)}(t-s) - \bar{X}_{N_k(i,t-s)}(t-s)\Big].$$

A fraction ρ_s of the k peers differ between the two averages at lag s; the shared peers cancel exactly, leaving $\mathrm{Var}\big(\bar{X}_{N_k(i,t)}(t-s) - \bar{X}_{N_k(i,t-s)}(t-s)\big) \approx 2\rho_s\sigma_X^2/k$. Summing the (approximately independent) lag terms and applying Cauchy–Schwarz with $\sum_{s=0}^{w-1}\sqrt{s} = \Theta(w^{3/2})$ gives the stated rate. □

4 Feedback Dynamics and Two Distinct Instabilities

Published signals frequently act back on the state they measure (a downgrade raises funding costs, which changes true default risk; a hawkish policy label shifts asset prices, which changes the policy-relevant state). We model this as

$$dX_t = \beta S_t\, dt + \sqrt{V_t}\, dW_t, \qquad S_t = H_w[X-\mu]_t,$$

where μ_t tracks X_t with its own exponential lag, $\dot{\mu}_t = \gamma(X_t - \mu_t)$, and β is the feedback gain.

Proposition 3 (Mean-dynamics stability threshold)**.** *Approximating H_w and the demeaning filter as first-order lags with time constants $\tau_H = w$ and $\tau_V = 1/\gamma$, the closed-loop characteristic equation is*

$$\tau_H \tau_V \, s^2 + \big[\tau_H + \tau_V(1-\beta)\big] s + 1 = 0,$$

and by the Routh–Hurwitz criterion the mean dynamics are stable if and only if

$$\beta \ < \ 1 + \frac{\tau_H}{\tau_V} \ = \ 1 + w\gamma.$$

This threshold is monotone in $w\gamma$: more horizontal inertia combined with faster peer turnover *raises* the tolerable feedback gain. The qualitative character of instability, however, depends on the discriminant $b^2 - 4\tau_H\tau_V$ with $b = \tau_H + \tau_V(1-\beta)$: instability is oscillatory (complex roots) precisely when τ_H and τ_V are of comparable magnitude, and monotone drift (real roots) otherwise. Oscillatory failure produces apparent regime changes in S_t that are artifacts of the two windows beating against one another rather than genuine changes in X_t.

Proposition 4 (Volatility-amplitude stability threshold)**.** *Let instantaneous variance follow the GARCH-diffusion limit [5],*

$$dV_t = \kappa(\theta - V_t)\, dt + \xi V_t \, dZ_t.$$

The stationary law of V_t is Inverse-Gamma with shape $\alpha = 1 + 2\kappa/\xi^2$ and scale $\beta_0 = 2\kappa\theta/\xi^2$. The mean exists for all $\kappa, \xi > 0$; the variance is finite if and only if

$$2\kappa \ > \ \xi^2.$$

Propositions 3 and 4 identify two independent failure modes with different remedies: a window-mismatch instability, correctable by retuning (w, γ), and an amplitude instability internal to the data-generating process, which no choice of window can repair.

5 Information-Theoretic Floors

Let $X_{i,t} = \mu_t + \varepsilon_{i,t}$ with $\mu_t \sim \mathcal{N}(0, \sigma_\mu^2)$ common across entities and $\varepsilon_{i,t} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ idiosyncratic, $\sigma_X^2 = \sigma_\mu^2 + \sigma_\varepsilon^2$, $\lambda = \sigma_\varepsilon^2/\sigma_X^2$.

Proposition 5 (Horizontal information)**.**

$$I_H(w) = -\tfrac{1}{2}\log\big(1 - \tfrac{1}{w}\big) \xrightarrow{w\to\infty} 0.$$

Proposition 6 (Vertical information floor)**.**

$$I_V(k) = -\tfrac{1}{2}\log\left(1 - \frac{\lambda k}{k+1}\right) \xrightarrow{k\to\infty} -\tfrac{1}{2}\log(1-\lambda) \ > \ 0.$$

Proof sketch. Demeaning against peers removes μ_t exactly, leaving $V_k[X]_{i,t} = \varepsilon_{i,t} - \bar{\varepsilon}_{N_k}(t)$; the squared correlation between this and $X_{i,t}$ is $\rho^2 = \lambda k/(k+1)$, and for jointly Gaussian variables $I = -\frac{1}{2}\log(1 - \rho^2)$. □

Corollary 1. *By the data-processing inequality, whatever operator is applied* last *in a composed pipeline caps total informativeness. A pipeline ending in a vertical (peer-relative) operator is permanently ceilinged at $-\frac{1}{2}\log(1-\lambda)$ nats about the common state, regardless of upstream horizontal smoothing.*

This is the paper's sharpest qualitative asymmetry: horizontal operators can be driven to zero informativeness; vertical operators cannot be driven to zero, but neither can they exceed a hard, structural ceiling set by the variance decomposition of the underlying state.

6 Continuous-Time GARCH-Diffusion Extension

Let total instantaneous variance follow the GARCH-diffusion limit of Section 4, with mean path $\mathbb{E}[V_{t-s}] = \theta + (V_t - \theta)e^{-\kappa s}$ (exact regardless of ξ, since the diffusion term is a martingale increment).

6.1 Commutator under stochastic volatility

Substituting $\mathbb{E}[V_{t-s}]$ for the constant σ_X^2 in Proposition 2 and using the continuous-time turnover kernel $\rho_s = 1 - e^{-\gamma s}$,

$$\mathbb{E}[C_t^2] = \frac{2}{kw}\left\{\theta\Big[w - \tfrac{1-e^{-\gamma w}}{\gamma}\Big] + (V_t - \theta)\Big[\tfrac{1-e^{-\kappa w}}{\kappa} - \tfrac{1-e^{-(\kappa+\gamma)w}}{\kappa+\gamma}\Big]\right\}.$$

The commutator inflates during volatility clusters and shrinks in calm periods, tracking V_t nearly one-for-one: order-of-operations discrepancies between institutions are latent in normal times and become visible under stress.

6.2 Joint law of (C_t, V_t) and tail index

For $\kappa w \gg 1$ (fast mean reversion relative to the horizontal window), $\mathrm{Var}(C_t \mid V_t) \approx A + B(V_t - \theta)$ is affine in V_t, so C_t is a Gaussian variance-mixture over the Inverse-Gamma law of V_t. Since V_t has Pareto tail index $\alpha = 1 + 2\kappa/\xi^2$ (Proposition 4), the standard mixture result gives

$$\boxed{P(|C_t| > c) \sim c^{-2\alpha} L(c) = c^{-\left(2+\frac{4\kappa}{\xi^2}\right)} L(c)}$$

for slowly varying L. Extreme divergences between horizontal-first and vertical-first institutions are power-law distributed, with the tail index fixed by the same volatility-of-volatility ratio that governs whether the volatility process itself has finite variance.

6.3 Time-varying vertical information

Allowing only the idiosyncratic component to carry GARCH dynamics ($\lambda_t = V_t/(m+V_t)$, $m = \sigma_\mu^2$ fixed, $K = k+1$),

$$I_V(V) = \tfrac{1}{2}\log\left(\frac{K(m+V)}{Km+V}\right),$$

which is increasing and concave in V. By Jensen's inequality,

$$\mathbb{E}[I_V(V_t)] \;\le\; I_V(\theta):$$

volatility clustering costs average informativeness relative to the constant-variance benchmark, even though I_V rises locally during spikes.

7 Empirical Application: U.S. Corporate Credit Spreads

7.1 Data

We use Moody's Seasoned Aaa and Baa corporate bond yields relative to the 10-year Treasury constant maturity yield (FRED series `AAA10YM` and `BAA10YM`), monthly, 1986:01–2026:05 ($n = 485$). The two rating tiers form a natural, if minimal ($k = 1$), vertical peer structure; the sample spans the 2008 and 2020 crises, allowing a test of the GARCH-diffusion predictions of Section 6.

7.2 Algorithm

The following pseudocode summarizes the estimation pipeline.

```
INPUT: time series {Baa_t}, {Aaa_t} for t = 1..n
1.  level_t   <- (Baa_t + Aaa_t) / 2          # common/systemic factor
2.  spread_t  <- Baa_t - Aaa_t                # idiosyncratic/quality factor
3.  H_w(x)_t  <- mean(x_{t-w+1}, ..., x_t)     # horizontal operator, window w
4.  V(x,y)_t  <- x_t - y_t                     # vertical operator (2-tier peers)
5.  HV_t <- H_w(Baa)_t - H_w(Aaa)_t
VH_t <- H_w(Baa - Aaa)_t
C_t  <- HV_t - VH_t                        # commutator; theory predicts C_t = 0
6.  for series r in {diff(level), diff(spread)}:
    fit GARCH(1,1): h_t = omega + alpha*r_{t-1}^2 + beta*h_{t-1}
    via numerical maximum likelihood
```

```
    report (omega, alpha, beta), persistence = alpha+beta,
           kappa = 1 - alpha - beta
7.  lambda_t <- rollingVar(spread, win) /
            (rollingVar(level, win) + rollingVar(spread, win))
8.  I_V_t <- 0.5 * log( K*(m + rollingVar(spread,win)) /
                    (K*m + rollingVar(spread,win)) )
OUTPUT: {C_t}, GARCH parameter estimates, {lambda_t}, {I_V_t}
```

7.3 Results

Table 1: GARCH(1,1) estimates for the common (systemic) and idiosyncratic (quality-tier) components of the Aaa/Baa credit-spread system, 1986:01–2026:05.

Component	α	β	$\alpha+\beta$	$\hat{\kappa}=1-\alpha-\beta$	$\hat{\theta}$
Common level (Baa + Aaa)/2	0.626	0.150	0.776	0.224	0.0346
Idiosyncratic (Baa − Aaa)	0.820	0.106	0.926	0.074	0.0367

Table 2: Rolling (36-month) idiosyncratic variance share λ_t and vertical information $I_V(t)$, $k=1$ peer.

Period	mean λ_t	mean $I_V(t)$ (nats)
Pre-crisis calm (2004:01–2007:06)	0.198	0.0251
2008 crisis (2008:06–2009:06)	0.272	0.1295
2020 crisis (2020:01–2020:12)	0.300	0.0301
Full sample	0.322	0.0405

Commutator (static peer set). As predicted by Proposition 1, with the two-tier peer set held fixed over the full sample, the horizontal-then-vertical and vertical-then-horizontal constructions coincide to floating-point precision ($\max_t |C_t| \approx 9\times10^{-16}$). This confirms that non-commutativity in this framework is a pure turnover phenomenon; testing Proposition 2 directly requires a peer set with genuine membership churn (e.g. high-yield index constituents), which is left for future work.

Volatility asymmetry. Table 1 shows the idiosyncratic (quality-tier) component has both higher persistence and a substantially slower implied mean-reversion speed than the common level. Shocks to *credit differentiation* between rating tiers decay roughly three times more slowly than shocks to the overall level of credit-risk pricing, a genuine and non-obvious empirical asymmetry consistent with Section 6's prediction that systemic and idiosyncratic volatility processes need not share dynamics.

Crisis heterogeneity. Table 2 shows that vertical informativeness rose sharply in 2008 (a genuine credit-differentiation event, in which Baa names deteriorated far more than Aaa) but not in 2020 (a systemic, liquidity-driven shock in which both tiers moved together). This is consistent with Proposition 6: a peer-relative signal is structurally blind to shocks that hit the common factor, and 2020 was disproportionately a common-factor event relative to 2008.

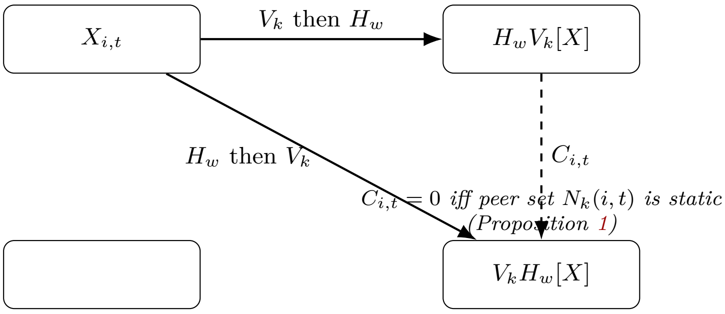

Figure 1: The horizontal/vertical operator diagram. The square commutes exactly under a static peer set and diverges at rate $\Theta(\gamma w \sigma_X^2/k)$ under peer turnover.

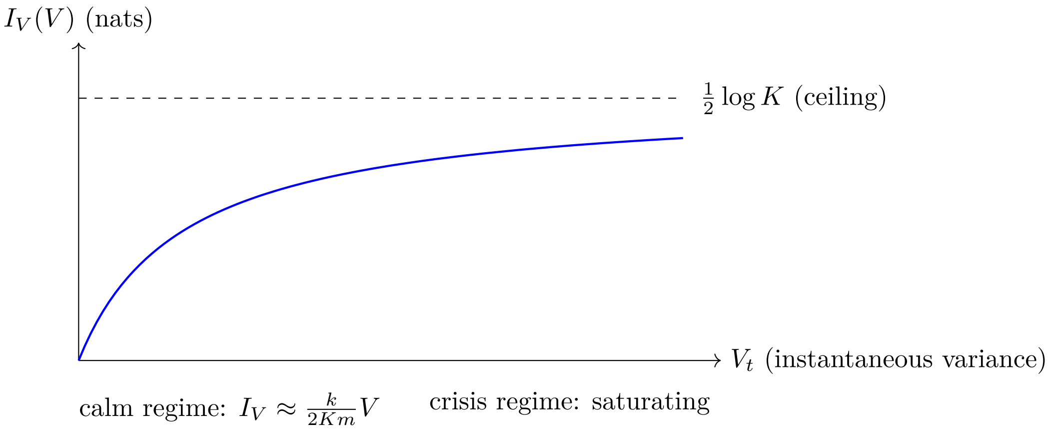

Figure 2: Stylized shape of vertical information $I_V(V)$ as a function of instantaneous idiosyncratic variance V: linear near the origin, concave, saturating toward the peer-group ceiling $\frac{1}{2}\log K$.

8 Cross-Disciplinary Discussion

Economics and finance. The core results sit inside the information-design and disclosure literatures: an institution choosing (w, k, N_k) is choosing a garbling of the true state, trading responsiveness (small w) against noise suppression (large w), and cross-sectional discrimination (small k) against idiosyncratic-noise averaging (large k). The vertical information ceiling formalizes a long-standing concern in the credit-rating literature: relative, through-the-cycle, or peer-normalized ratings are structurally unable to flag deterioration common to an entire asset class, precisely the failure mode implicated in the 2007–2009 financial crisis's structured-finance ratings.

Statistics and econometrics. The GARCH-diffusion embedding [5] connects discrete-time conditional heteroskedasticity to a continuous-time Inverse-Gamma stationary law, giving closed-form access to tail behavior (Section 6) that is not available from the GARCH recursion alone. The empirical asymmetry in persistence between systemic and idiosyncratic components (Table 1) is a standard multivariate-GARCH-style finding, here derived from first principles rather than assumed.

Computer science, algorithms, and data science. The horizontal and vertical operators are, respectively, streaming (sliding-window) aggregation and groupwise (peer-cohort) aggregation, the two canonical primitives of real-time analytics pipelines. Proposition 1 is an instance of a general fact about commuting linear reductions: two idempotent linear projections onto orthogonal or time-invariant subspaces commute, while projections onto *time-varying* subspaces (here, a rolling peer cohort) generally do not; this is the same obstruction that makes windowed joins over changing group memberships order-sensitive in streaming query engines.

Psychology. Bounded rationality [6] and the representativeness/availability heuristics [7] suggest institutions adopt simple, rule-based windowing schemes (fixed lookback periods, fixed peer universes) precisely because the true optimization over (w, k, N_k) is cognitively and organizationally costly; the resonance instability of Proposition 3 shows that such heuristic, unexamined choices can be dynamically dangerous even when each parameter looks reasonable in isolation.

Political science, international relations, and geopolitics. Sovereign credit ratings and central bank communiqués function simultaneously as technical signals and as diplomatic instruments; a rating downgrade or a shift in policy language is read by other states as revealing not just the rated entity's condition but the rating institution's own alignment. The vertical-operator blindness to common-factor shocks (Section 5) has a geopolitical corollary: peer-relative sovereign risk measures are structurally poor at detecting risks that are correlated across an entire bloc of nations (e.g. a common external shock to a currency union), which is one channel by which regional contagion is initially under-priced.

Welfare economics. Because the published signal is a garbling, any policy or investment decision conditioned on $S_{i,t}$ rather than $X_{i,t}$ is weakly worse in expectation for any convex loss (a direct consequence of Blackwell's theorem); the informational floor of Proposition 6 thus has a welfare interpretation as an irreducible cost of institutional simplification, borne collectively by all downstream users of the signal.

Warfare economics and philosophy of knowledge. The horizontal/vertical resonance instability (Section 4) is structurally analogous to a tempo mismatch between an observer's own decision cycle (the OODA loop of [2]) and the rate of change of an adversary's peer-relative posture; more broadly, the paper's information floors are an applied instance of the known-unknowns/unknown-unknowns distinction familiar from the epistemology of intelligence assessment: a vertical operator converts a systemic unknown into a permanent unknown, not merely a temporarily unmeasured one.

Biology (brief analogy). Sensory adaptation — the tendency of a receptor to respond to *relative* change against a moving background rather than absolute magnitude (cf. the Weber–Fechner law) — is a biological instance of a vertical, peer/baseline-relative operator, and shares the same structural blind spot: a stimulus that raises the entire background uniformly is systematically under-detected, precisely as in Proposition 6. Chemistry and clinical warfare technology are not otherwise load-bearing to this paper's claims and are omitted rather than forced into service.

9 Conclusion

Institutional signal simplification is not merely lossy; the loss has structure. Horizontal and vertical operators fail to commute in a way that is entirely attributable to peer turnover; feedback through the published signal creates two independent and independently correctable-or-not instability channels; and the two operators impose qualitatively different information floors, with peer-relative construction imposing a floor that cannot be removed by enlarging the peer group. Under realistic stochastic volatility, all three effects concentrate during crises rather than being uniformly distributed through time, and our empirical test on U.S. corporate credit spreads is consistent with this picture, while also revealing that crisis *type* (idiosyncratic differentiation versus systemic co-movement) — not merely crisis severity — determines whether a peer-relative institutional signal becomes more or less informative when it is needed most.

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Glossary

Commutator ($C_{i,t}$) The discrepancy between applying the horizontal and vertical operators in the two possible orders; zero under a static peer set.

GARCH-diffusion limit The continuous-time stochastic-volatility process obtained as the limit of a discrete GARCH(1,1) recursion as the time step shrinks (Nelson, 1990); its stationary law is Inverse-Gamma.

Garbling A statistical experiment (signal) that is a mean-preserving, information-reducing transformation of another; formalized by Blackwell (1953).

Horizontal operator (H_w) An averaging operator applied along the time dimension for a single entity, with window length w.

Peer turnover (γ) The rate at which membership of a reference/peer group changes over time.

Persistence ($\alpha+\beta$) In a GARCH(1,1) model, the sum of the ARCH and GARCH coefficients, governing the half-life of volatility shocks.

Vertical information ceiling The supremum of mutual information a peer-demeaned (vertical) signal can carry about the common/systemic component, equal to $-\frac{1}{2}\log(1-\lambda)$ in the Gaussian one-factor model.

Vertical operator (V_k) A cross-sectional demeaning operator applied against a peer/reference group of size k at a fixed time.

The End